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INVITED PAPER

Hybrid cognitive diagnostic model

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Abstract

Cognitive diagnostic models can be classified into two categories based on the type of interaction between attributes: disjunctive and conjunctive. A representative example of the former is the "Deterministic Input Noisy-Or gate" (DINO) model, and of the latter is the "Deterministic Input Noisy-And gate" (DINA) model. However, fixing the interaction form to be either disjunctive or conjunctive may be based on a strong assumption. Therefore, we developed a new hybrid cognitive diagnostic model in which the item response function is represented as a weighted mixture of disjunctive and conjunctive item response functions. This made it possible to estimate the quantitative degree of each interaction type for each item, while keeping the parameters within reasonable limits. The proposed model was formalized as a Bayesian model and estimated using the Hamiltonian Monte Carlo algorithm. A Monte Carlo simulation confirmed adequate parameter recovery of the proposed method. In an empirical application to actual mathematics test data, the proposed model achieved better predictive performance than the DINA and DINO models. The obtained posteriors of the mixture weights were found to be heterogeneous among items, indicating key advantages of the proposed approach.

Keywords Cognitive diagnostic models · DINA model · DINO model · Attribute interactions

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1 Introduction

In the educational and psychological sciences, there is growing demand for detailed assessments of students' skills, cognitive processes, and problem-solving strategies (collectively called *attributes*) within achievement contexts. For this purpose, cognitive diagnostic models (CDMs; e.g., de la Torre and Minchen 2014), which are a specific class of discrete latent variable models, have been developed for assessing each student's mastery (or lack thereof) of attributes based on properly designed test data. By applying these models, the mastery profiles of pre-defined cognitive subtasks can be estimated, which can be used to capture each student's cognitive strengths and weaknesses. This information can then be used to tailor instructions for optimizing student learning (de la Torre 2009).

Existing CDMs can be classified into two major categories based on the assumption of how the attributes interact to produce an observed item response: *disjunctive* and *conjunctive* (DiBelloet al. 2006; Rupp et al. 2010). The difference between these two models lies in the mathematical form of how latent variables that correspond to different attributes are combined. Disjunctive models assume that low levels of one attribute can be compensated for with high levels of another. Conversely, conjunctive models assume that a student must have all required attributes to correctly answer each item.

Among many CDMs that have been proposed so far, the most representative conjunctive CDM is the deterministic input noisy-and gate (DINA; e.g., Junker and Sijtsma 2001) model, and its disjunctive alternative is the deterministic input noisy-or gate (DINO; Templin and Henson 2006) model. Both are formulated using simple, interpretable terms. The fact that the parameters can be interpreted in a straightforward manner makes these models attractive (Liu et al. 2009; de la Torre and Lee 2010). To empirically apply the CDMs, researchers normally need to determine either the appropriate disjunctive or conjunctive approach for modeling the test data of interest. Literature has emphasized that to do this appropriately, a comprehensive and substantive understanding of the interactive nature of attributes is essential (DiBello et al. 2006).

Still, this choice between the two modeling approaches is often difficult to make. For example, in a study that made diagnostic inferences regarding English language reading skills, Jang (2009) revealed that a lack of vocabulary was not easily compensated for with comprehension skills. Conversely, Jang also noted that the process of applying vocabulary knowledge varied depending on a student's language background, prior knowledge, and degree of context dependency, which could lead different students to use varying and mutually disjunctive strategies to answer the same items. Li et al. (2016) discussed theoretical evidence to support both the disjunctive and conjunctive nature of cognitive skills; therefore, the interaction form between attributes that is better for testing reading comprehension is unclear. Similar discussions have also occurred in diagnostic tests in mathematics: based on a detailed validation process, Tjoe and de la Torre (2014) developed a test in which students were expected to take multiple steps to solve

problems, with mathematical attributes assumed for each step. This implies that if a student solves the problem in the way the question preparer anticipated, they possess all the required attributes and conjunctively combine them to correctly answer it. However, for mathematics test items, often, there are multiple strategies to solve any question (e.g., Huo and de la Torre 2014; Ma and Guo 2019). Furthermore, students can apply attributes in an unexpected manner. Hence, it is possible that, even for this kind of diagnostic test, and at least for some items, disjunctive models represent the response process better than conjunctive ones. Previous empirical model comparison studies also support the potential difficulty in determining the interaction form between attributes (Park and Cho 2017; Yamaguchi and Okada 2018).

To mitigate the problem of choosing either a disjunctive or conjunctive model, one might move away from applying a constrained model, and instead use a saturated and unconstrained model form of CDM, which have been proposed in the general frameworks of the generalized DINA (G-DINA; de la Torre 2011), log-linear CDM (LCDM; Henson et al. 2009), and general cognitive diagnostic model (GDM; von Davier 2008). However, as Ravand (2016) has pointed out with reference to Rojas et al. (2012), the parameters in these general models are often hard to interpret in practice. This is because the saturated models are overparameterized: they include all possible interactions between the attributes in addition to the main effects (de la Torre 2011). Adequate parameter constraints are essential for obtaining interpretable and meaningful insights from the model, which are especially important in educational and psychological applications to fulfill the need for accountability.

Accordingly, the central idea pursued in the present study is to keep the item response functions (IRFs) as simple as possible, while explicitly considering the uncertainty between the disjunctive and conjunctive decision strategies. The prior mixed empirical findings, which we reviewed earlier, may be partly due to actual tests not conforming completely to either a disjunctive or conjunctive model. Thus, we developed a hybrid CDM (H-CDM), in which the probability of answering correctly on a particular item is based on a mixture of disjunctive and conjunctive components. Thus, our model considers that each category has a unique parameter that determines the *degree* of disjunctive (as opposed to conjunctive) interactions between attributes. Formal parameterization of the model as well as interpretation of the new parameter will be provided in the next section. Statistically, this idea is implemented in the model as a Bayesian mixture of disjunctive and conjunctive model components. More specifically, we considered the mixture of the DINA and DINO-type IRFs, due to their popularity and parsimony, which makes them suitable for extending into a mixture model. Although the Bayesian approach has been popular in CDM framework (e.g., Culpepper 2015; Zhan et al. 2019a, b), the authors are not aware of any Bayesian mixture formulation of disjunctive and conjunctive item response functions, which we propose in this paper.

The remainder of the paper is structured as follows. First, we introduce the DINA and DINO models, and use them to formulate the proposed H-CDM based on a Bayesian framework. Next, we conducted a Monte Carlo simulation study that shows adequate parameter recovery performance of the Hamiltonian Monte Carlo estimation. Next, we applied the proposed H-CDM to real mathematics test data and compared its

predictive performance with existing DINA and DINO models. Estimation results are then interpreted. Finally, we summarize the proposal, discuss it in relation to existing generalized CDM frameworks, and suggest future research directions.

2 Model

2.1 DINA and DINO Models

The IRFs of the DINA and DINO models were used to construct the IRF for the proposed H-CDM. Let x_{ij} be the observed binary item response of student $i(= 1, \ldots, I)$ to item $j(= 1, \ldots, J)$, with $x_{ij} = 1$ if the student answers the item correctly and $x_{ij} = 0$ if incorrect. The IRF of the DINA model is defined as

$$\mathrm{P}_{\mathrm{DINA}}\left(x_{ij} = 1 | s_j^A, g_j^A, \eta_{ij}\right) = \left(1 - s_j^A\right)^{\eta_{ij}} \left(g_j^A\right)^{1-\eta_{ij}}, \tag{1}$$

where the slip parameter s_j^A and the guessing parameter g_j^A are given in the conditional form as

$$s_j^A = \mathrm{P}\left(x_{ij} = 0 | \eta_{ij} = 1\right), \tag{2}$$

$$g_j^A = \mathrm{P}\left(x_{ij} = 1 | \eta_{ij} = 0\right), \tag{3}$$

and η_{ij} is the ideal response, which indicates whether the student is intrinsically capable of providing a correct response to the item. This ideal response is a function of requisites from the item and the student's skills. The former is represented by the Q-matrix (Tatsuoka 1983), the element of which is a binary variable q_{jk} that equals 1 if the attribute k is required to correctly answer item j and 0 otherwise. Similarly, the latter is represented by the attribute mastery pattern matrix, the element of which is also a binary variable α_{ik} that equals 1 if the attribute k is mastered by student i and 0 otherwise. The ideal response η_{ij} in the DINA model is defined as

$$\eta_{ij} = \prod_{k=1}^{K} \alpha_{ik}^{q_{jk}}. \tag{4}$$

This ideal response equals 1 when student i possesses all the required attributes for item j and 0 when s/he misses at least one required attribute. This means that the DINA model classifies students into two groups for each item: those who mastered all the required attributes and those who missed at least one required attribute. Note that we arbitrarily define $0^0 = 1$ in this paper.

The IRF of the DINO model is virtually the same as the DINA model. Namely, it is defined as

$$\mathrm{P}_{\mathrm{DINO}}\left(x_{ij} = 1 | s_j^O, g_j^O, \omega_{ij}\right) = \left(1 - s_j^O\right)^{\omega_{ij}} \left(g_j^O\right)^{1-\omega_{ij}}. \tag{5}$$

The difference between the DINA and DINO models lies in the functional form of the ideal response. The ideal response in the DINO model, denoted by ω_{ij}, is defined as

$$\omega_{ij} = 1 - \prod_{k=1}^{K} \left(1 - \alpha_{ik}\right)^{q_{jk}}. \tag{6}$$

This ideal response equals 1 when student i possesses at least one required attribute for item j and 0 when s/he does not possess any required attributes. This means that the DINO model classifies students into the following two groups for each item: those who mastered at least one required attribute and those who missed all the required attributes. The slip and guessing parameters are the same as Eqs. (2) and (3), except for the notational difference of the ideal response parameter.

2.2 Formulation of the hybrid CDM

The IRF of the proposed hybrid CDM is defined as a weighted mixture of the DINA-type and DINO-type IRFs. Specifically, for items that have more than one required attribute (i.e., $\sum_k q_{jk} > 1$), the IRF is given as

$$\begin{aligned} &\mathrm{P}\left(x_{ij} = 1 | s_j^A, g_j^A, s_j^O, g_j^O, \delta_j, \eta_{ij}, \omega_{ij}\right) \\ &= \delta_j \mathrm{P}_{\mathrm{DINA}}\left(x_{ij} = 1 | s_j^A, g_j^A, \omega_{ij}\right) + \left(1 - \delta_j\right)\mathrm{P}_{\mathrm{DINO}}\left(x_{ij} = 1 | s_j^O, g_j^O, \omega_{ij}\right) \\ &= \delta_j \left(1 - s_j^A\right)^{\eta_{ij}} \left(g_j^A\right)^{1-\eta_{ij}} + \left(1 - \delta_j\right)\left(1 - s_j^O\right)^{\omega_{ij}} \left(g_j^O\right)^{1-\omega_{ij}}. \end{aligned} \tag{7}$$

Here, the newly introduced mixing parameter δ_j, which is a value between 0 and 1, represents the weight given to the DINA-type IRF as opposed to the DINO-type IRF. If δ_j is close to 1, then the item j should behave like the IRF of the DINA model. On the other hand, when it is close to 0, item j should behave like the IRF of the DINO model.

Because the proposed IRF is a mixture of two components, its interpretation can be provided in a similar manner as finite mixture models. An obvious situation where this two-component mixture model is directly applicable is when the response for item j is drawn from a population that consists of two groups of students, each having DINA-type and DINO-type IRFs, respectively, in proportions δ_j and $\left(1 - \delta_j\right)$ (McLachlan et al. 2019, p. 358; McLachlan and Peel 2000, p. 7). In the current study, we mainly employ this interpretation of the weight parameter. However, other interpretations of the weight parameter are also possible. For instance, in another situation, the mixture is thought of as a convenient parsimonious representation of an IRF that requires greater flexibility than the standard DINA- or DINO-type IRFs (Richardson and Green 1997, p.733). In this context, the parameter may be understood as the relative disjunctivity of item j because it is the weight of the DINA-type component (as opposed to the DINO-type) for representing the item response.

Other parameters in Eq. (7) are the same as those in the DINA and DINO models. The superscripts "A" and "O" distinguish the slip and guessing parameters of the DINA or DINO models, respectively. We refer to s_j^A and g_j^A as the DINA-type parameters and s_j^O and g_j^O as the DINO-type parameters.

Equation (7) is defined for those items that require more than one attribute. In other words, the hybrid nature of the proposed IRF is meaningful only when items require more than one attribute. In contrast, for items that require only one attribute, the ideal responses of the DINA-type and DINO-type IRFs, which are η_{ij} (Eq. 4) and ω_{ij} (Eq. 6), reduce to the same form because there is no "interaction between attributes." Therefore, for those items, we only estimate the slip and guessing parameters, which can be interpreted in the context of both the DINA and DINO models, and omit the mixing parameter δ_j. For those parameters, without loss of generality, we omit the superscript A or O in the slip and guessing parameters and denote the ideal response by η_{ij}.

Based on the above, the complete IRF for the proposed model can be written as

$$\begin{cases} \mathrm{P}\left(x_{ij} = 1 | s_j^A, g_j^A, s_j^O, g_j^O, \delta_j, \eta_{ij}, \omega_{ij}\right) & \text{if } \sum_k q_{jk} > 1, \\ \mathrm{P}\left(x_{ij} = 1 | s_j^A, g_j^A, \eta_{ij}\right) & \text{if } \sum_k q_{jk} = 1. \end{cases} \tag{8}$$

2.3 Bayesian estimation

When estimating parameters for Eq. (8), we adopted a fully Bayesian formulation. This allowed for formal and coherent treatment of uncertainty, and the resultant Bayesian model could be estimated using the general Markov chain Monte Carlo (MCMC) algorithm (Bernardo and Smith 2000; Gelman et al. 2013). As the total possible patterns for the binary element of the attribute profile vector $\boldsymbol{\alpha}_i = \left[\alpha_{i1}, \ldots, \alpha_{ik}, \ldots, \alpha_{iK}\right]'$ is 2^K, the CDM can be considered a latent class model. Thus, the probability of the latent attribute configuration $\boldsymbol{\alpha}_c$, which is shared by all students in latent class c, is represented as $\nu_c = \mathrm{P}(\boldsymbol{\alpha}_i = \boldsymbol{\alpha}_c | \boldsymbol{\nu})$, where $\boldsymbol{\nu} = \left[\nu_1, \ldots, \nu_{2^K}\right]'$ is the vector of the latent class probability of length 2^K. Under the assumption of conditional independence, the likelihood of observing an $\mathbf{X}$ matrix whose element is x_{ij}, when $\sum_k q_{jk} > 1$, is given as

$$\begin{aligned} \mathrm{P}(\mathbf{X}|\boldsymbol{s}, \boldsymbol{g}, \boldsymbol{\nu}, \boldsymbol{\delta}) = \prod_{i=1}^{I} \sum_{\boldsymbol{\alpha}_c \in [0,1]^K} \nu_c \prod_{j=1}^{J} & \left[\delta_j \left(1 - s_j^A\right)^{\eta_{cj}} \left(g_j^A\right)^{(1-\eta_{cj})} + (1-\delta_j)\left(1 - s_j^O\right)^{\omega_{cj}} \left(g_j^O\right)^{(1-\omega_{cj})}\right]^{x_{ij}} \\ & \left[\delta_j \left(s_j^A\right)^{\eta_{cj}} \left(1 - g_j^A\right)^{(1-\eta_{cj})} + (1-\delta_j)\left(s_j^O\right)^{\omega_{cj}} \left(1 - g_j^O\right)^{(1-\omega_{cj})}\right]^{1-x_{ij}}, \end{aligned} \tag{9}$$

where η_{cj} represents the ideal response of class c in the DINA-type IRF, which is given as $\eta_{cj} = \prod_{k=1}^{K} \alpha_{ck}^{q_{jk}}$, and similarly $\omega_{cj} = 1 - \prod_{k=1}^{K} \left(1 - \alpha_{ck}\right)^{q_{jk}}$ represents the corresponding ideal response in the DINO-type IRF. $\boldsymbol{s}$ and $\boldsymbol{g}$ represent the vector of slip and guessing parameters, respectively. In the same manner, the likelihood when $\sum_k q_{jk} = 1$ is given as

$$\mathrm{P}(\mathbf{X}|\boldsymbol{s},\boldsymbol{g},\boldsymbol{\nu},\boldsymbol{\delta}) = \prod_{i=1}^{I} \sum_{\boldsymbol{\alpha}_c \in [0,1]^K} \nu_c \prod_{j=1}^{J} \left[\left(1-s_j\right)^{\eta_{cj}} \left(g_j\right)^{(1-\eta_{cj})} \right]^{x_{ij}} \left[\left(s_j\right)^{\eta_{cj}} \left(1-g_j\right)^{(1-\eta_{cj})} \right]^{1-x_{ij}}, \tag{10}$$

which is the same form as the original likelihood of the DINA model and also the same as the DINO model. We used these form of likelihood functions, rather than that of Eq. (8), for ease of estimation.

Junker and Sijtsma (2001) established that the sum of the slip and guessing parameters should be less than 1. To satisfy this condition, we set uniform prior distribution over the support that satisfies the monotonicity constraint $0 \leq g_j < 1 - s_j \leq 1$ for all DINA and DINO item parameters:

$$\left(s_j^A, g_j^A\right) \sim \mathrm{Uniform}(0,1)^2_{\mathcal{I}(0 \leq g_j^A < 1 - s_j^A \leq 1)}, \tag{11}$$

$$\left(s_j^O, g_j^O\right) \sim \mathrm{Uniform}(0,1)^2_{\mathcal{I}\left(0 \leq g_j^O < 1 - s_j^O \leq 1\right)}, \tag{12}$$

which also apply in the case when $\sum_k q_{jk} = 1$. For other parameters, the following natural priors can be considered:

$$\boldsymbol{\nu} \sim \mathrm{Dirichlet}(1), \tag{13}$$

$$\delta_j \sim \mathrm{Uniform}(0,1). \tag{14}$$

The above model can be estimated using Stan (Carpenter et al. 2017), which is a Bayesian modeling language that implements an adaptive Hamiltonian Monte Carlo algorithm (HMC; e.g., Hoffman and Gelman 2014). This algorithm overcomes some of the problems of a conventional MCMC algorithm within complex latent variable models. This enabled us to effectively generate samples from the posterior distribution. The Stan code used for implementing the model can be found in the Online Supplemental Material (https://osf.io/z53mw/).

3 Simulation study

3.1 Simulation settings

We conducted a simulation study to check parameter recovery performance of the proposed model. Detailed simulation settings are described below. In this simulation, we considered two conditions for item parameters, two conditions for attribute mastery patterns, and two conditions of test length. Specifically, in the high-quality item condition, all item parameters s_j^A, g_j^A, s_j^O, and g_j^O were generated from $\mathrm{Uniform}(0.05, 0.20)$, independently for each item and ordered to satisfy monotonicity. In contrast, in the low-quality item condition, they were generated from $\mathrm{Uniform}(0.25, 0.35)$. We followed Culpepper (2015) for the conditions of true

Table 1 Q-matrix used in the simulation study for the short test length condition

Item	Attribute				Item	Attribute			
	1	2	3	4		1	2	3	4
1	1	0	0	0	10	1	0	1	0
2	0	1	0	0	11	1	0	0	1
3	0	0	1	0	12	0	1	1	0
4	0	0	0	1	13	0	1	0	1
5	1	0	0	0	14	0	0	1	1
6	0	1	0	0	15	1	1	1	0
7	0	0	1	0	16	1	1	0	1
8	0	0	0	1	17	1	0	1	1
9	1	1	0	0	18	0	1	1	1

attribute mastery pattern probabilities ν_c. Specifically, in the flat distribution condition, the probability ν_c was kept constant at $1/2^K$ for all c. On the other hand, in the non-flat distribution condition, the class with three to four attribute masteries were twice as likely as the class with zero to two attribute masteries. Furthermore, in a similar manner to Zhan et al. (2018), the test length condition was either short (18 items) or long (36 items). The Q-matrix used in the short test length condition is shown in Table 1. This Q-matrix has eight items that require only one attribute, five items that require two attributes, and four items that require three attributes to answer correctly. The long test length condition employed a double stacking of this Q-matrix. Thus, the Q-matrix satisfies an identifiability condition presented by Chen et al. (2015). It is important for this identifiability condition to hold because, otherwise, the parameters cannot be uniquely estimated. Parameter δ_j was independently generated from Uniform(0.20, 0.80). We set the sample size to $I = 500$ and the number of attributes at 4. Using these true parameter and attribute values, we generated item response data using Eq. (8).

The parameters were estimated using the model described in the previous section. The rstan() function of the "RStan" package (Stan Development Team 2018) in R was utilized. We generated four independent chains of MCMC samples. Initial values were randomly generated. The number of iterations was 2,000, and the first half were discarded as burn-in. The posterior means and 95% credible intervals were calculated from the second half of the samples. Convergence of the chains was checked with $\hat{R}$ of the Gelman–Rubin diagnostic (Gelman and Rubin 1992). The entire simulation process was repeated 30 times. R and Stan code used in this simulation can be found in our Open Science Framework project webpage at https://osf.io/z53mw/.

To evaluate parameter recovery, we calculated the bias and root mean squared error (RMSE) of the estimates, which are, respectively, calculated as

$$\text{Bias} = \frac{1}{30}\sum_{m=1}^{30}\left(\hat{\theta}_m - \theta_{\text{true}}\right), \tag{15}$$

$$\mathrm{RMSE} = \sqrt{\frac{1}{30}\sum_{m=1}^{30}\left(\hat{\theta}_m - \theta_{\mathrm{true}}\right)^2}, \tag{16}$$

where $\hat{\theta}_m$ represents the parameter estimate obtained from the m-th replication in the simulation, and θ_{true} represents the parameter's true value.

3.2 Results

$\hat{R}$ s for all parameters, in all replications, were below 1.10, suggesting that the MCMC chains converged. Bias and RMSE values for all item parameters in the short test length condition are presented in Figs. 1 and 2. Overall, these values were adequately small for all four conditions, indicating good parameter recovery performance. Additionally, biases for items that required only one attribute (items 1–8) tended to be smaller than those that required more than one attribute (items 9–18). A similar tendency was also found for the RMSE values. The bias and RMSE for the DINO-type item parameters were slightly larger than those of the DINA-type ones, but these values were still adequately small in general. The bias for the mixing parameter tended to fluctuate in some conditions, which might be due to sampling errors resulting from the limited number of replications. The RMSE values in low-quality test condition were larger than the values in high-quality test conditions. Similarly, the Bias and RMSE values for all item parameters in the long test length condition are presented in Figs. 3 and 4. The Bias and RMSE values showed similar tendencies as the short test length condition. The bias values were adequately small for all parameters. The RMSE values of the DINA-type parameters for items that measure more than one attribute (items 9–18 and 27–36) were relatively larger. The RMSEs of the DINO-type parameters were all around 0.05. The variability of the mixing parameter estimates in the low-quality item conditions were again relatively larger than that of other parameters. However, in general, parameter recovery was satisfactory even under the long test length condition. In summary, the H-CDM parameters were adequately recovered by the adaptive HMC estimation.

4 Application to real data

4.1 Data and analysis settings

The proposed H-CDM was applied to the fourth-grade Trends in International Mathematics and Science Study 2007 (TIMSS; Foy and Olson 2009) dataset[1] for the United States, which has been analyzed in previous studies (Lee et al. 2011; Park and Lee 2014; Yamaguchi and Okada 2018). We analyzed the participants without

[1] The dataset is available at the TIMSS website (https://timssandpirls.bc.edu/TIMSS2007/idb_ug.html) together with the item information (https://timssandpirls.bc.edu/TIMSS2007/PDF/T07_Items.zip).

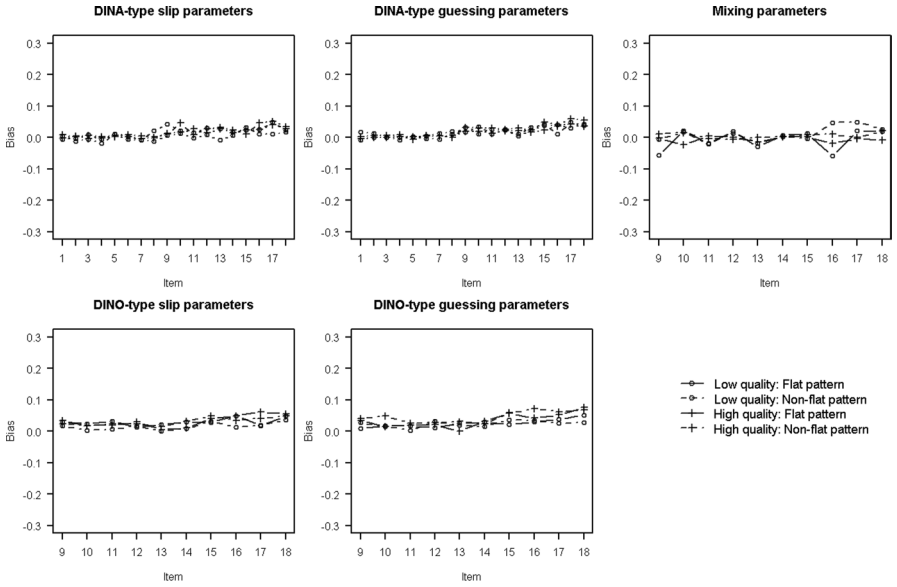

Fig. 1 Bias of item and mixing parameters under the four simulation conditions in the short test length condition

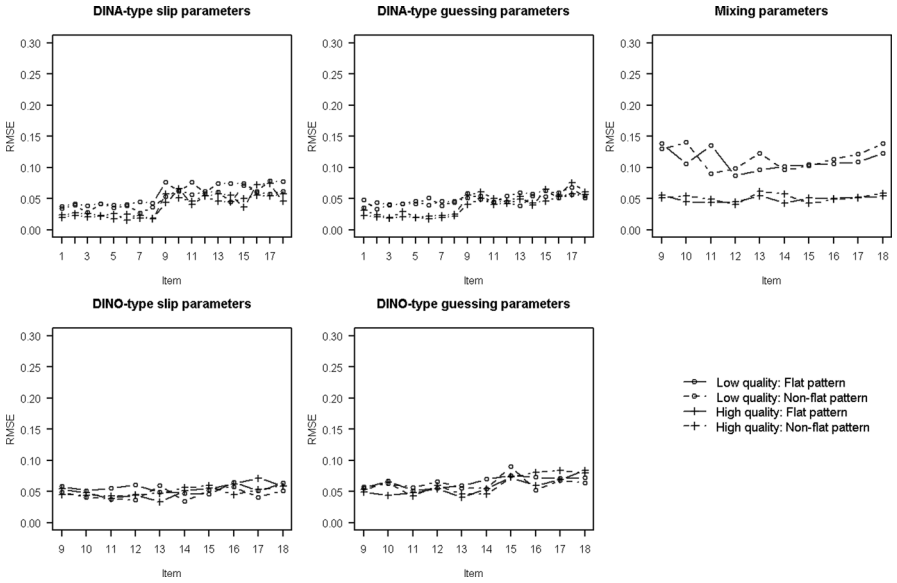

Fig. 2 RMSE of item and mixing parameters under the four simulation conditions in the short test length condition

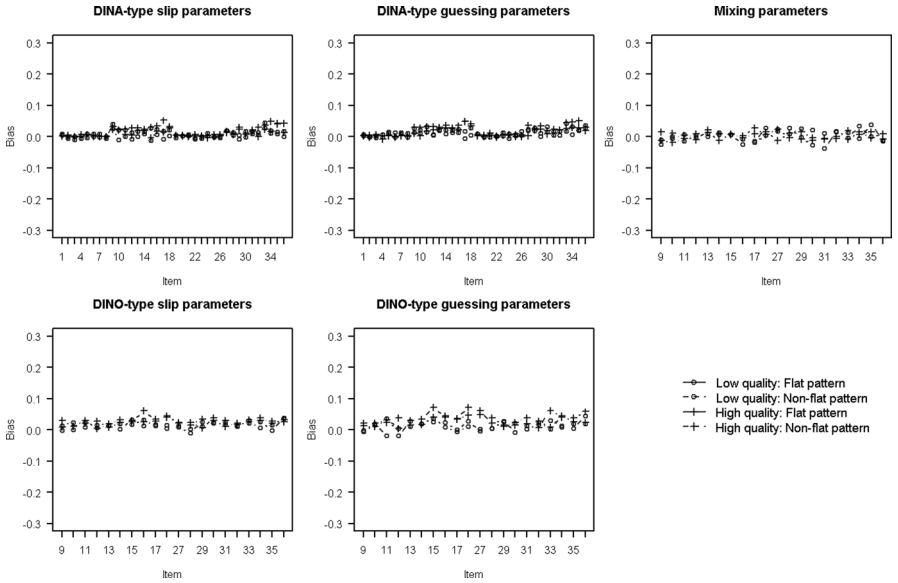

Fig. 3 Bias of item and mixing parameters under the four simulation conditions in the long test length condition

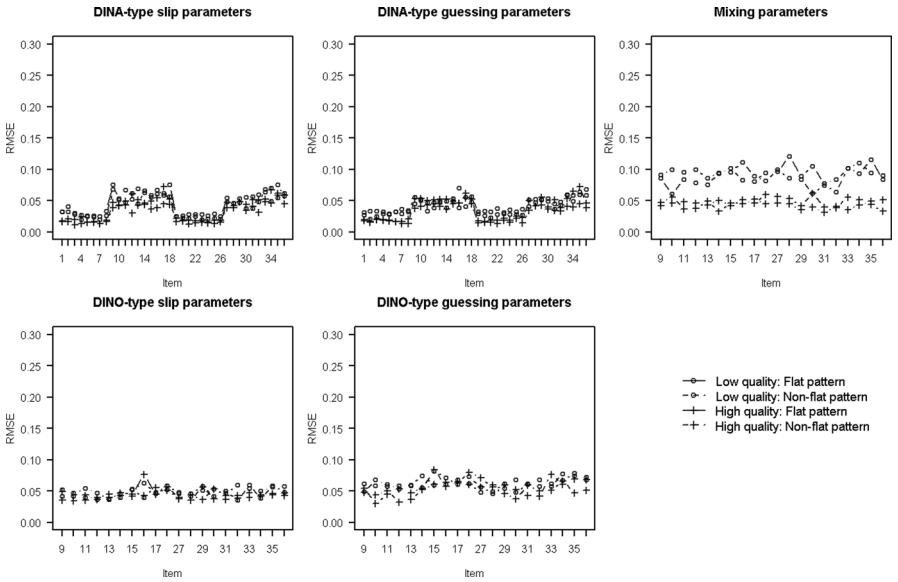

Fig. 4 RMSE of item and mixing parameters under the four simulation conditions in the long test length condition

missing observations, and the sample size was 395. This 25-item dataset is suitable for our purposes because not only the observed responses but also the original items used in the tests are available. Hence, we can interpret the obtained model parameters based on the actual item content and context. For this dataset, Lee et al. (2011, Table 2) proposed a Q-matrix that consists of 15 attributes from the three content domains of "Number," "Geometric Shapes and Measurement," and "Data and Display." Thereafter, Park and Lee (2014) proposed a simplified version of the Q-matrix that consists of seven attributes: "Whole Numbers" (N1), "Fractions and Decimals" (N2), "Number Sentences, Patterns, and Relationships" (N3), "Lines and Angles" (G4), "Two- and Three-Dimensional Shapes" (G5), "Location and Movement" (G6), and "Reading, Interpreting, Organizing, and Representing" (D7). We used this

Table 2 Q-matrix for the TIMSS 2007 mathematics fourth grade data (Park and Lee 2014)

Item	Attribute						
	N1	N2	N3	G4	G5	G6	D7
M041052	1	0	0	0	0	0	0
M041056	0	1	0	0	0	0	0
M041069	1	1	0	0	0	0	0
M041076	1	1	0	0	0	0	0
M041281	1	0	1	0	0	0	0
M041164	0	0	0	0	1	1	0
M041146	0	0	0	1	1	1	0
M041152	1	0	0	0	1	0	0
M041258A	0	0	0	0	1	0	0
M041258B	0	0	0	1	1	0	0
M041131	1	0	0	1	0	0	0
M041275	1	0	0	0	0	0	1
M041186	1	0	0	0	0	0	1
M041336	1	1	0	0	0	0	1
M031303	1	0	0	0	0	0	0
M031309	1	0	0	0	0	0	0
M031245	1	0	1	0	0	0	0
M031242A	1	0	1	0	0	0	0
M031242B	1	0	0	0	0	0	1
M031242C	1	0	1	0	0	0	1
M031247	1	0	1	0	0	0	0
M031219	0	0	0	0	1	1	0
M031173	1	0	0	0	0	0	0
M031085	0	0	0	0	1	0	0
M031172	1	0	0	0	0	0	1

N1 whole numbers; *N2* fractions and decimals; *N3* number sentences, patterns, and relationships; *G4* lines and angles; *G5* two- and three-dimensional shapes; *G6* location and movement; *D7* reading, interpreting, organizing, and representing

Q-matrix in the current analysis. For example, the third item (M041069)— "Which fraction is equal to 2/3?"—measured attributes N1 and N2. This means that knowledge of whole numbers, fractions, and decimals is relevant in correctly answering this item. The choice options were A:3/4, B:4/9, C:4/6, and D:3/2, and the correct answer was C.

The data were analyzed using the proposed H-CDM, as well as the DINA and DINO models, to compare relative model fit. All models were implemented in Stan; the R and Stan code used in this application can be found in our Open Science Framework project webpage at https://osf.io/z53mw/. The number of MCMC chains was three, and the number of iterations was 1,500. The first half of the iterations were discarded as burn-in. Convergence of the chains was checked by $\hat{R}$.

For model comparison, we used the widely applicable information criterion (WAIC; Watanabe, 2010). The WAIC is a fully Bayesian value that uses the entire posterior distribution and is an asymptotically unbiased estimate of the out-of-sample prediction error. A smaller WAIC indicates better model fit (Vehtari et al. 2017). Additionally, WAIC can be easily computed based on the MCMC samples. In this study, the waic() function in the "loo" package (Vehtari et al. 2017) was used for calculating WAIC.

4.2 Results

All $\hat{R}$ s in all models were smaller than 1.10, which suggests that the MCMC chains were converged. Obtained traces and empirical posteriors also indicated good mixing and convergence of the samples. Figure 5 shows the trace plots and empirical posterior distributions of item parameters for an item (Item M041281, for illustrative purposes).

The resultant WAIC values for the H-CDM, DINA model, and DINO model were 10,407.789 ($SE = 100.101$), 10,555.466 ($SE = 102.366$), and 12,626.394 ($SE = 124.540$), respectively. These results indicate that the proposed H-CDM was

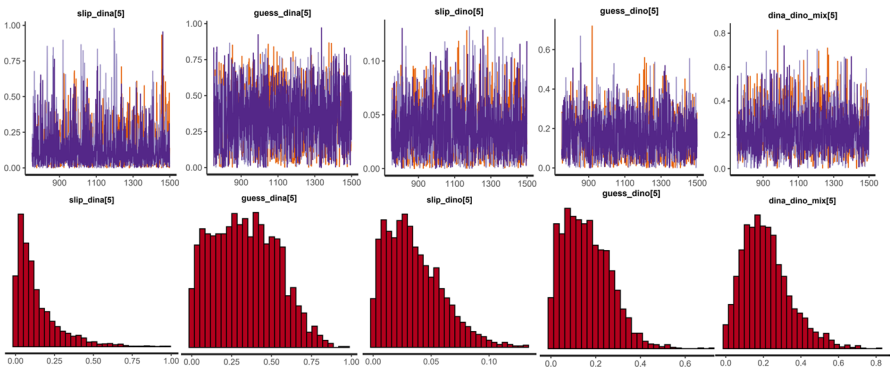

Fig. 5 Trace plots (upper row) and empirical posterior distributions (lower row) for the item parameters of item 5 (M041281). The left two panels represent the slip and guessing parameters of the DINA portion; the next two panels represent those of the DINO portion; the rightmost panel represents the mixed parameter

the best model in terms of WAIC. Thus, we examined the resultant parameter estimates for the H-CDM in further detail.

The item and mixing parameter estimates are shown in Table 3. The shape of the posterior distributions of the mixing parameters varied between items, and accordingly, the posterior means also varied between close to 0 and close to 1. For example, the posterior means of the mixing parameters were 0.217 [95% CI = (0.021, 0.527)] for item M041281 and 0.142 [95% CI = (0.004, 0.518)] for item M041186. Thus, we can infer that many students employed conjunctive rather than disjunctive combinations of attributes for these items. Note that, as illustrated in the leftmost density of Fig. 5 for item M041281, posterior modes were, in general, closer to the bounds than the posterior means for [0, 1]-bounded parameter such as this one. In contrast, the posterior means were 0.807 [95% CI = (0.558, 0.991)] for item M041336 and 0.862 [95% CI = (0.579, 0.996)] for item M031245. Thus, we can infer that many students employed disjunctive rather than conjunctive combinations of attributes for these items. Furthermore, the posterior means were close to 0.5 for some other items. For example, they were 0.520 [95% CI = (0.121, 0.935)] for item M041152 and 0.522 [95% CI = (0.193, 0.932)] for item M031242C. For this type of item, students as a whole may not use one of the disjunctive or conjunctive combinations of attributes. Rather, they apparently have different combination mechanisms depending on the individual or the occasion. Below, we further elaborate on the empirical interpretations of the results for the above-mentioned items.

Item M041281, which measured attributes N1 (Whole Numbers) and N3 (Number Sentences, Patterns and Relationships), had a large weight on the disjunctive component of the model. The statement of the item was "Layne had 32 pencils and 4 boxes for the pencils. He put the same number of pencils into each box. Which number sentence describes how many pencils are put into the box?" and its choice options were A: $32+4$, B: 32–4, C: 32×4, and D: $32\div4$, with D as the correct answer. Based on the weight estimates from the H-CDM, for most students, mastery of one of the two attributes can be sufficient to answer this item correctly. This makes sense for the following reason. The attribute N1 includes the skill "Recognize multiples, computing with whole numbers using the four operations, and estimating computations" and "Solve problems, including those set in real life contexts" (Lee et al. 2011, Table 2; the following definitions for the attributes in quotation marks are also cited from the same table). From the problem statement, we can see that they are exactly the skills required to solve item M041281. Likewise, the attribute N3 includes the skill "Describe relationships in patterns and their extensions; generate pairs of whole numbers by a given rule and identify a rule for every relationship given pairs of whole numbers," which, again, corresponds to the context of this item because the problem involves a pair of numbers. In this way, our estimation results, in which a student with high skill in either of the two attributes is likely to answer this item correctly, can be supported by empirical considerations.

Item M041186, which measured attributes N1 and D7, was another item that had a large weight on the disjunctive component. The attribute D7 is defined as a combination of the skills "Read the data from tables, pictographs, bar graphs, and pie charts," "Comparing and understanding how to use information from data," and "Understanding different representations and organizing data using tables,

Table 3 Item Parameter estimates of the TIMSS 2007 mathematics fourth grade data

Item	#Required attributes	Parameter	Posterior mean	95% CI
M041052	1	Slip	0.091	[0.056, 0.134]
M041056	1		0.205	[0.111, 0.299]
M041258A	1		0.040	[0.003, 0.089]
M031303	1		0.100	[0.057, 0.149]
M031309	1		0.120	[0.059, 0.188]
M031173	1		0.104	[0.057, 0.156]
M031085	1		0.326	[0.234, 0.409]
M041069	2	Slip (DINA-type)	0.284	[0.015, 0.645]
M041076	2		0.166	[0.009, 0.520]
M041281	2		0.141	[0.004, 0.557]
M041164	2		0.178	[0.002, 0.724]
M041146	3		0.092	[0.002, 0.401]
M041152	2		0.266	[0.016, 0.624]
M041258B	2		0.710	[0.076, 0.953]
M041131	2		0.457	[0.049, 0.779]
M041275	2		0.086	[0.003, 0.315]
M041186	2		0.303	[0.013, 0.814]
M041336	3		0.141	[0.012, 0.288]
M031245	2		0.307	[0.070, 0.465]
M031242A	2		0.128	[0.005, 0.545]
M031242B	2		0.136	[0.008, 0.337]
M031242C	3		0.121	[0.008, 0.340]
M031247	2		0.588	[0.082, 0.798]
M031219	2		0.387	[0.025, 0.799]
M031172	2		0.174	[0.004, 0.671]
M041069	2	Slip (DINO-type)	0.247	[0.017, 0.454]
M041076	2		0.101	[0.005, 0.256]
M041281	2		0.035	[0.002, 0.092]
M041164	2		0.018	[0.000, 0.075]
M041146	3		0.107	[0.003, 0.378]
M041152	2		0.271	[0.014, 0.616]
M041258B	2		0.653	[0.039, 0.956]
M041131	2		0.487	[0.037, 0.865]
M041275	2		0.193	[0.007, 0.724]
M041186	2		0.071	[0.008, 0.136]
M041336	3		0.331	[0.015, 0.798]
M031245	2		0.469	[0.024, 0.909]
M031242A	2		0.088	[0.004, 0.290]
M031242B	2		0.204	[0.011, 0.477]
M031242C	3		0.134	[0.007, 0.409]
M031247	2		0.573	[0.043, 0.930]
M031219	2		0.396	[0.022, 0.798]
M031172	2		0.035	[0.002, 0.099]

Table 3 (continued)

Item	#Required attributes	Parameter	Posterior mean	95% CI
M041069	2	Mixing	0.421	[0.098, 0.733]
M041076	2		0.372	[0.055, 0.769]
M041281	2		0.217	[0.021, 0.527]
M041164	2		0.158	[0.002, 0.826]
M041146	3		0.539	[0.185, 0.964]
M041152	2		0.520	[0.121, 0.935]
M041258B	2		0.598	[0.014, 0.991]
M041131	2		0.584	[0.092, 0.969]
M041275	2		0.665	[0.152, 0.991]
M041186	2		0.142	[0.004, 0.518]
M041336	3		0.807	[0.558, 0.991]
M031245	2		0.862	[0.579, 0.996]
M031242A	2		0.430	[0.055, 0.925]
M031242B	2		0.635	[0.387, 0.903]
M031242C	3		0.522	[0.193, 0.932]
M031247	2		0.724	[0.221, 0.991]
M031219	2		0.517	[0.046, 0.974
M031172	2		0.209	[0.009, 0.633]
M041052	1	Guessing	0.747	[0.670, 0.815]
M041056	1		0.057	[0.003, 0.141]
M041258A	1		0.665	[0.566, 0.750]
M031303	1		0.594	[0.504, 0.676]
M031309	1		0.241	[0.160, 0.328]
M031173	1		0.389	[0.293, 0.486]
M031085	1		0.196	[0.096, 0.301]
M041069	2	Guessing (DINA-type)	0.246	[0.015, 0.545]
M041076	2		0.465	[0.028, 0.813]
M041281	2		0.334	[0.015, 0.730]
M041164	2		0.557	[0.029, 0.976]
M041146	3		0.246	[0.014, 0.512]
M041152	2		0.283	[0.015, 0.598]
M041258B	2		0.088	[0.002, 0.506]
M041131	2		0.245	[0.022, 0.567]
M041275	2		0.623	[0.082, 0.843]
M041186	2		0.503	[0.030, 0.924]
M041336	3		0.199	[0.019, 0.327]
M031245	2		0.068	[0.006, 0.140]
M031242A	2		0.534	[0.044, 0.860]
M031242B	2		0.113	[0.005, 0.263]
M031242C	3		0.419	[0.029, 0.685]
M031247	2		0.081	[0.005, 0.228]
M031219	2		0.403	[0.028, 0.798]
M031172	2		0.509	[0.028, 0.911]

Table 3 (continued)

Item	#Required attributes	Parameter	Posterior mean	95% CI
M041069	2	Guessing (DINO-type)	0.198	[0.015, 0.455]
M041076	2		0.422	[0.042, 0.716]
M041281	2		0.161	[0.009, 0.389]
M041164	2		0.705	[0.208, 0.885]
M041146	3		0.263	[0.007, 0.800]
M041152	2		0.280	[0.017, 0.674]
M041258B	2		0.149	[0.002, 0.700]
M041131	2		0.285	[0.017, 0.738]
M041275	2		0.569	[0.030, 0.941]
M041186	2		0.407	[0.083, 0.619]
M041336	3		0.385	[0.021, 0.850]
M031245	2		0.291	[0.012, 0.810]
M031242A	2		0.514	[0.044, 0.855]
M031242B	2		0.181	[0.007, 0.577]
M031242C	3		0.423	[0.028, 0.844]
M031247	2		0.214	[0.008, 0.749]
M031219	2		0.411	[0.036, 0.826]
M031172	2		0.431	[0.064, 0.682]

pictographs, and bar graphs." In summary, this attribute corresponds to the skills of using tables and graphs. This item includes a graph that consisted of some apple marks for 4 days, with a side note that each mark represents 10 apples. Specifically, seven and a half marks were shown for Monday, five for Tuesday, one for Wednesday, and half for Thursday. The question was: "On which day did John pick five apples?" The choice options were A: Monday, B: Tuesday, C: Wednesday, and D: Thursday, and the correct answer was D. Based on our H-CDM estimation results, again, mastery in one of the two attributes was sufficient to answer this item correctly. This makes sense for the following reasons. A student who had high skill in Whole Numbers (N1) could compute the numbers that corresponds to each day, for example, by computing $7.5 \times 10 = 75$ for Monday, and achieve the correct answer. Meanwhile, a student who has high skill in Data & Display (D7) could take a different approach to the correct answer: they could read the image with the side information that one mark corresponds to 10 apples, and choose the correct day that had half a mark through visual reasoning. In this way, our estimation results indicated the two distinct and warranted approaches, and either one or both of these can be employed by many students to provide the correct answer for this item.

In contrast, item M041336, which measured the attributes N1, N2, and D7, had a large weight on the conjunctive component. The text of this item described two classes, Class A and B, each having 40 students. A pie chart and a bar plot accompanied the text. The pie chart displayed the ratio between boys and girls in class A.

The bar plot indicated the number of boys and girls in class B. The question was "There are more girls in Class A. How many more?" To correctly answer this item, a student must calculate the number of girls in Class A using the pie chart, read the number of girls in Class B from the bar plot, and then make a subtraction. In this way, this item requires combinatory use of different skills for most students, and a failure in one component would likely result in a wrong answer. The obtained estimate of the mixing parameter can be interpreted as a reflection of such conjunctive use of attributes.

As another example, M031245, which measured the attributes N1 and N3, also had a large weight on the conjunctive component. These two attributes were the same as those measured by item M041281, which had a large effect on the disjunctive component. The text of the item was "$12 \div 3 = \blacksquare \div 2$. In this number sentence, what number does ■ stand for?" The choice options were A: 2, B: 4, C: 6, and D: 8, and the correct answer was D. To answer this item, not just the mastery of whole numbers, but also of finding a missing number in the equation by converting a fraction into multiplication by the reciprocal is required for most students. This requirement exactly corresponds to one of the skills of the attribute N3: "Find the missing number or operation and model simple situations involving unknowns in number sentence or expressions." We can see that, compared to Item M041281, the cognitive process required to solve the current item is more complicated, and hence, a more conjunctive combination of attributes is required.

Lastly, item M041152, which measured the attributes N1 and G5, had a weight close to 0.5, suggesting the existence of two groups of about the same size, each of which employ disjunctive and conjunctive combinations. The attribute G5 involved skills regarding the classification and calculation of geometric figures and objects. In this item, a picture of a rectangular fence was presented together with its height (3 m) and width (4 m) information. The text of the item was "Patrick is painting one side of a fence. The fence is 4 m long and 3 m high. What is the area that Patrick has to paint?" For this item, there was a duality of information: the size of the fence was presented both in the text as well as in the figure. Therefore, some students could disjunctively utilize one of them to reach the correct answer, using the number skill (N1) in the case of a text-based approach or using the geometric skill (G5) in the case of a figure-based approach. Meanwhile, other students could conjunctively combine both the text and figure information to find the correct answer. In this way, the estimation result of the weight parameter close to 0.5 would reflect the fact that the response mechanism cannot be explained by a sole-disjunctive nor a sole-conjunctive combination mechanism; in this case, it may reflect the possible co-existence of both disjunctive and conjunctive approaches among students.

As seen above, the proposed H-CDM is more flexible than the parsimonious DINA and DINO models, had better fit in terms of information criterion, and could express the groups that employ different ways of combining attributes to solve each item. The estimation results of the proposed H-CDM were interpretable in the context of the items, and provided possibilities to explain the underlying cognitive processes employed by the students for answering questions.

5 Discussion

The present study was conducted to develop an H-CDM in which the IRF is given as a mixture of a DINA IRF (which represents disjunctive interactions between attributes) and a DINO IRF (which represents conjunctive interactions). This model allowed us to estimate the relative degree of disjunctive and conjunctive interactions, instead of fixing the interaction form a priori, while restricting the number of parameters.

The simulation study demonstrated that Bayesian estimation using an adaptive HMC algorithm adequately recovered the parameter values. Furthermore, an example using real data revealed the added value of the proposed H-CDM as opposed to existing DINA and DINO models, both in terms of predictive errors and flexibly representing the degree of compensation for each item.

As discussed, one of the H-CDM's main strengths is its ability to flexibly take into account both the disjunctive and conjunctive nature of interactions between attributes, while preserving the good interpretability of the model parameters. Previous studies have discussed that DINA and DINO models, albeit widely used, might be too restricted for empirical application (Henson et al. 2009; Li et al. 2016; Yamaguchi and Okada 2018). These models may be too parsimonious to model students' cognitive processes in that they assume either disjunctive or conjunctive interactions between attributes. On the other hand, the parameters of the overparameterized generalized models, such as the higher-order interaction terms, tend to be difficult to interpret and communicate. This might prevent these models from educational and psychological applications in practice. The proposed H-CDM addresses this problem by mixing DINA and DINO IRFs; this provides more flexibility in modeling a cognitive process and simplifies data interpretations, while keeping the number of parameters as small as possible.

In the real-data application, the proposed H-CDM was found to be the best performing model for the TIMSS 2007 fourth grade mathematics dataset in terms of WAIC. According to its estimation results, the proportion of students who employ disjunctive (as opposed to conjunctive) attribute combinations varied between the items. Even when items measure the same set of attributes, it would be reasonable to think that respondents' cognitive processes in combining them to answer questions may differ depending on the items. The mixing parameters of the H-CDM capture and reflect such differences: it would come close to 0 or 1 when one of the combination mechanisms dominates in the sample, but close to 0.5 when there are heterogeneities. From this standpoint, a typical application approach that selects one of the disjunctive or conjunctive models for the whole test dataset can be a rather strong assumption. The proposed H-CDM flexibly represents heterogeneous proportions of disjunctive use of attributes among items while maintaining the parsimony and interpretability of the model component. We believe that this is a strength of the H-CDM.

It is important to note that the term mixture model is used for two types of somewhat different contexts. One is the case when each mixture component has the same functional form, and the other is when each has a different functional form (Tijmstra et al. 2018). Clearly, this study considers the latter case, which is sometimes called the "latent mixture model" (Lee and Wagenmakers 2013). In the former type of

models, Bayesian estimation with MCMC algorithm requires special care due to the label switching problem caused by the nonidentifiability of the components (Jasra et al. 2005; Okada and Lee 2016). In contrast, this is not a problem in the latter case thanks to the different functional form of each component. This type of Bayesian mixture model provides a principled approach to estimating the different contributions, as well as the uncertainties, of each mixture component.

One potential issue with the proposed model is its estimation time. The MCMC estimation of the proposed model generally required a much longer computation time as compared to the simpler DINA and DINO models. This is one reason that the current simulation study was limited in its number of conditions; improvement of computing capabilities with time would gradually resolve this issue. Furthermore, the computer code employed in our simulation is available from the Open Science Framework, and, hence, an empirical researcher can examine the model performance under the condition of interest.

Employing a faster approximation algorithm, such as a variational Bayes, could be an important direction for future studies. In fact, the proposed H-CDM could actually reflect a sub-model of the aforementioned generalized models. This possibility is worth further investigating to allow for the application of a faster, generalized estimation algorithm. Another possible future direction involves considering a mixture of other disjunctive and conjunctive model IRFs. This could provide even more flexibility, albeit at the cost of computational burden. Nevertheless, to model higher-order cognitive skills, such an extension could be pragmatically meaningful. In addition, methods to extend the proposed model to the computer adaptive testing format are required for its efficiency in diagnosing individual cognitive status. Furthermore, an empirical study that collects test data for the proposed H-CDM would be informative, as it would help us provide further understanding of the connection between item response process and the model parameters.

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Compliance with ethical standards

Conflict of interest We have no conflicts of interest to declare.

Appendix

Data analysis code is available in Supplementary Material (https://osf.io/z53mw/).

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